Mehraan Chowdhury

Sydney, NSW, Australia • mehraan.chowdhury@gmail.com • +61 416 810 665 •
github.com/mehraanc • linkedin.com/in/mehraanc

Full Australian work rights valid through April 2029 (Subclass 485 visa)

SUMMARY

Machine learning researcher with experience building and evaluating models for computer vision and natural language processing. Comfortable owning a problem end to end, from reviewing the literature and designing experiments, preparing data, adapting open source models, and training them from scratch or fine-tuning pre-trained weights. Works mainly in Python and PyTorch. Most recently developed novel methods at the University of Western Australia for screening obstructive sleep apnea from 3D facial images, producing results that outperformed previously published work, published as two first-author IEEE conference papers.

Holds a Master of Philosophy from the University of Western Australia, with a thesis ranked in the top 5%, and a Master of Information Technology (Artificial Intelligence) from the University of Melbourne with a WAM of 81.1 (H1 average). Also brings 3.5 years of front end development.

PROFESSIONAL EXPERIENCE

Machine Learning Researcher Sep 2023 - Mar 2026

University of Western Australia, Perth

- Developed novel methods for screening obstructive sleep apnea from 3D facial images, training machine learning and deep learning models to produce results that outperformed previously published work
- Reviewed the literature for deep learning techniques for classifying 3D meshes and point clouds, identifying high-performing methods that had not been applied to the relevant problem, and designed a program of experiments to test whether they improved classification performance
- Developed a dataset annotation process so medical experts with no technical background could label 3D scans to produce the ground-truth dataset
- Adapted two open-source deep learning codebases, a graph convolutional network and a multi-view CNN, to a new landmarking task, training both models from scratch to locate anatomically significant points on 3D faces
- Built and evaluated a range of machine learning classifiers, applying stratification, cross-validation, normalisation, standardisation, ablation studies, feature reduction, and feature importance analysis to produce results that outperformed previously published work
- Adapted two open-source generative pre-trained point cloud transformers for classification, training under various configurations with and without additional pre-training, both from scratch and from provided pre-trained weights to produce results that outperformed previously published work
- **Tools Used:** Python, PyTorch, scikit-learn, NumPy, Optuna, Open3D, trimesh, MeshLab, CloudCompare, NetworkX, Linux, Git, Cline, Claude Code

Machine Learning Researcher Jul 2022 - Jan 2023

University of Melbourne, Melbourne

- Built an NLP pipeline in Python to predict relevant programming context from Stack Overflow answers using transformer models (BERT) and NLTK
- Scraped and processed large volumes of developer Q&A data using Beautiful Soup and ChromeDriver

Front-End Web Developer Apr 2016 - Dec 2019

Revo Interactive Ltd, Dhaka, Bangladesh

- Converted design wireframes into responsive production user interfaces using HTML5, CSS3, JavaScript, and jQuery
- Ensured cross-browser consistency across Chrome and Firefox
- Integrated SEO content into HTML markup to improve search visibility for client websites
- Diagnosed and resolved bugs across the front-end codebase, reducing time-to-fix on reported issues
- Contributed design and development ideas during sprint planning, helping shape feature direction across the team

EDUCATION

Master of Philosophy (Computer Science) Sep 2023 - Mar 2026
University of Western Australia

- Thesis: *Deep Learning-Based Screening of Obstructive Sleep Apnea Using 3D Facial Images*
- Available in the UWA Research Repository
- Ranked in the top 5% of theses examined, as noted in the examiner report
- Two first-author IEEE conference publications produced from thesis research

Master of Information Technology (Artificial Intelligence) Mar 2021 - Dec 2022
University of Melbourne

- WAM: 81.1 (H1 / High Distinction average)
- Thesis: *Predicting Relevant Context for Programming Questions using Stack Overflow Answers*

Bachelor of Science (Neuroscience)
University of Melbourne Mar 2013 - Nov 2015

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, HTML5, CSS3, SQL, Bash
- **Machine Learning:** PyTorch, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Optuna, Hugging Face, NLTK, Jupyter
- **AI coding agents:** Claude Code, Cline
- **Computer Vision:** OpenCV, PIL, Open3D, trimesh, MeshLab, CloudCompare
- **Web automation:** Beautiful Soup, Selenium / ChromeDriver, Playwright
- **Tools & Methods:** Git, GitHub, Linux, LaTeX, Agile, Waterfall

PUBLICATIONS

- M. Chowdhury, M. Bennamoun, F. Boussaid, S. M. Shamsul Islam, "Classifying Obstructive Sleep Apnea from 3D Facial Images Using Automated Landmark Detection and Geodesic Distances," *2025 IEEE Conference on Future Machine Learning and Data Science (FMLDS)*, Los Angeles, 2025, pp. 602-607. doi: 10.1109/FMLDS67896.2025.00098
- M. Chowdhury, M. Bennamoun, F. Boussaid, S. M. Shamsul Islam, "Automatic Detection of Peripheral Facial Landmarks Using 3D Facial Data," *2024 IVCNZ Conference on Image and Vision Computing New Zealand*, Christchurch, 2024, pp. 1-6. doi: 10.1109/IVCNZ64857.2024.10794203

REFERENCES

Mohammed Bennamoun
Professor, School of Physics, Mathematics and Computing
University of Western Australia
mohammed.bennamoun@uwa.edu.au

Farid Boussaid
Professor, School of Engineering
University of Western Australia
farid.boussaid@uwa.edu.au